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DISEÑO DE UN MODELO DE TOKENIZACIÓN PARA UN FONDO DE PRIVATE EQUITY

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DESIGN OF A TOKENIZATION MODEL FOR A PRIVATE EQUITY FUND

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ABSTRACT

This study designs, specifies, and evaluates a digital infrastructure based on blockchain networks and smart contracts to tokenize a private equity fund, reducing administrative costs and enabling fund democratization and internal secondary market liquidity.

Keywords: tokenization, blockchain, smart contracts, private equity, DLT.

1. Introduction

When people think about investing, they almost automatically picture buying shares on the stock market or purchasing government bonds. However, this traditional perspective overlooks a compelling reality of the global business landscape: over 90% of companies with revenues exceeding 100 million USD are private (Hamilton Lane & Capital IQ, 2023). Access to the yield and growth of this ecosystem is managed through private equity (direct investment in non-listed companies to optimize operations, accelerate growth, and later divest at a higher value), an asset class with an extraordinary track record of returns (Harris et al., 2014). Even so, minimum tickets of hundreds of thousands of dollars and a ten-year lock-up render it practically invisible to retail investors. The cause lies in fund managers' analog infrastructures, designed to serve a very small investor base with large tickets. With their current friction and inefficiencies, it is impossible to open the fund to a larger pool of investors.

In parallel, blockchain technology is transforming finance: first with Bitcoin and later with smart contracts. Existing institutional initiatives have implemented blockchain as a distribution channel while leaving the fund's operational machinery untouched: the asset is tokenized, but the process is not. This project proposes going one step further and tokenizing and automating the fund's entire lifecycle, transforming the operational model that governs private equity to democratize this investment product.

The study is built around the following research question:

> How can a fund manager company in Spain design, specify, and evaluate a DLT and smart contract infrastructure that enables the real democratization of a private equity fund and provides viable internal secondary liquidity, without administrative bottlenecks or compromising CNMV regulatory requirements?

To answer it, the study adopts the Design Science Research methodology (Peffers et al., 2007), designing and evaluating the artifact, the tokenized fund's operational and technological infrastructure, across six sequential phases that map onto the chapters of this report.

2. System specification and design constraints

The reference model consists of launching a tokenized Spanish private equity fund with a target size of 150M €, aimed at attracting a critical mass of 1,500 investors, as opposed to the typical average of 20 institutional investors of the traditional model.

The design of the new operational model must comply with the following technical design constraints and data control specifications, derived from the applicable regulatory framework (Spanish Law 22/2014; Commission Delegated Regulation (EU) No. 231/2013):

- Network synchronization: the state of the blockchain must match the fund manager's register of partners in real time, through a bidirectional synchronization middleware that rules out any discrepancy.
- Privacy and pseudonymity: recording personal data on-chain is strictly prohibited; wallets are validated and managed through cryptographic credentials.
- Asset control and governance: the fund manager must retain administrative authority over token circulation.

3. Operational and technical infrastructure description

The tokenized fund model optimizes the operational lifecycle through three pillars:

1. Onboarding and digital identity: automation of the KYC flow through an identity provider that issues cryptographic credentials.
2. Automated mechanisms: smart contracts that automatically trigger token minting and automate all operational processes in the fund's lifecycle.
3. Secondary market liquidity: in addition to providing the secondary market portal, the fund steps in as buyer and seller of last resort through treasury stock operations programmed with automatic rules.

The infrastructure is designed to run on a corporate permissioned network, Hyperledger Besu, under QBFT consensus, ensuring zero transaction costs and transaction privacy.

The Solidity architecture is split into five modular smart contracts grouped into three logical layers (token, information, and operations) under ERC-3643 token standard (ERC3643 Association, 2023):

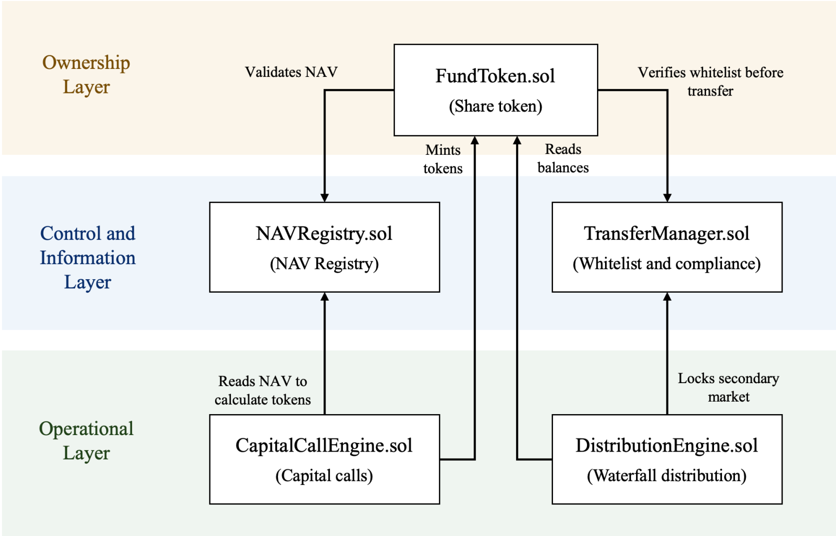

Figure 1 Modular smart contract architecture (author's elaboration)

To work around the EVM's lack of floating-point arithmetic, the contract embeds a Q64.64 binary fixed-point library that computes with accounting exactness.

4. Results

- The smart contract architecture was validated using a robust testing suite in Hardhat, comprising 190 unit tests to verify error limits and edge cases. In addition, 26 sequential integration checks were simulated to recreate real-world fund lifecycle scenarios.

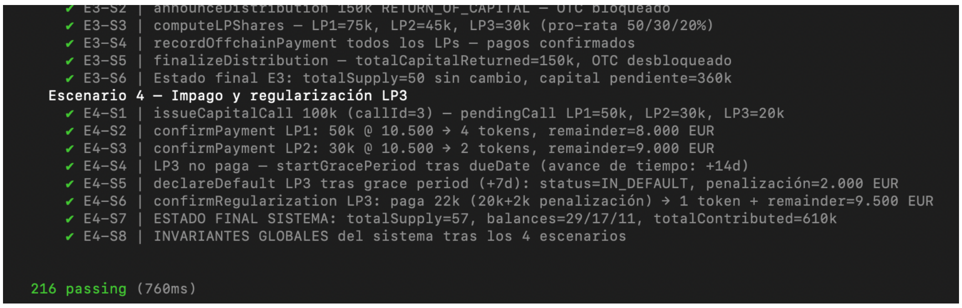

Figure 2 Consolidated console output after running the 216 tests

- The secondary market delivers measurable liquidity: against the 20-30% structural liquidity discount estimated without a secondary market (Franzoni et al., 2012), the model calibrates a base discount of 6.5%.
- The 10-year financial feasibility study, modeled from the general partner's perspective, shows solid profitability under base-case assumptions. With an initial investment of 1,193,470 €, the project yields an IRR of 28.85%, with a discounted payback period of 2.80 years.
- In execution times, administrative agents, data management, and operational expenses, the tokenized model represents the most efficient alternative to scale the fund: cuts operations from days to minutes and enables scaling to thousands of investors without friction, with a secondary market built on quarterly windows.

5. Conclusions and limitations

This project provides preliminary evidence of the technical, legal, and economic viability of employing DLT to modernize private equity in Spain, with a design reviewed by industry specialists. Although the stated objectives have been met, the system presents two fundamental limitations:

1. Regulatory constraints: the absence of specific regulation adapted to fund tokenization forces the retention of traditional figures (notarial structures, manual registers), which prevents moving the vehicle's entire governance onto the blockchain and lowering the minimum ticket any further.

2. Secondary market volume and adoption: despite implementing an automated secondary market, its success depends on market volume and depth (Nadauld et al., 2019); the lack of a critical mass of active investors and the misconception that equates tokenization with speculative crypto assets may create resistance to adoption.

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